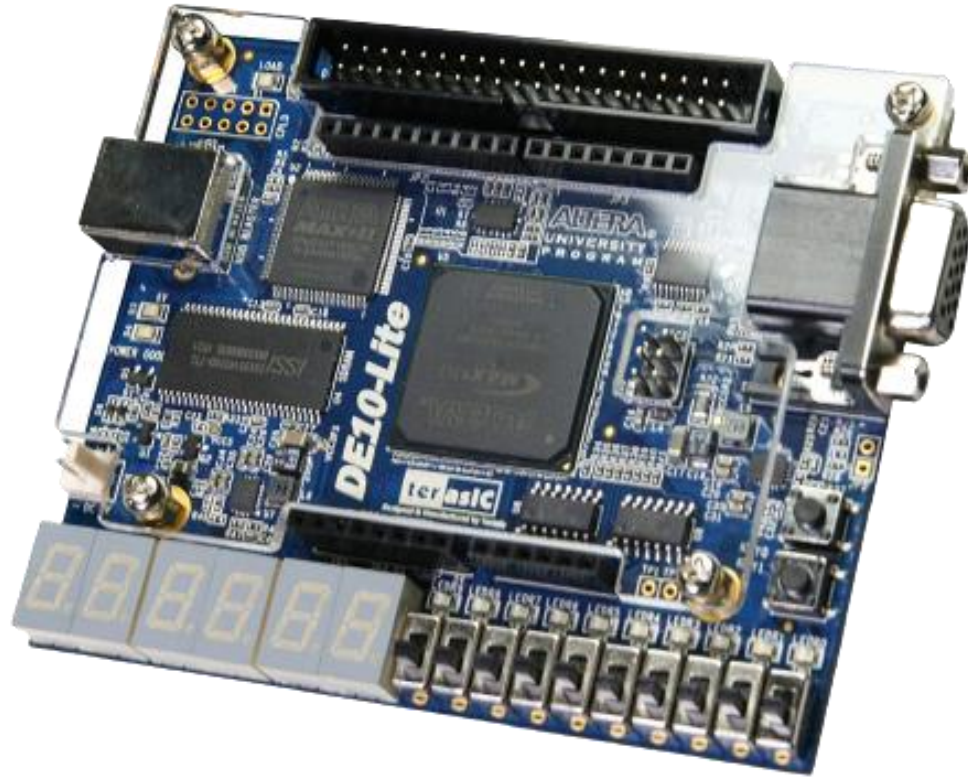


ROTEIRO PROJETO 6



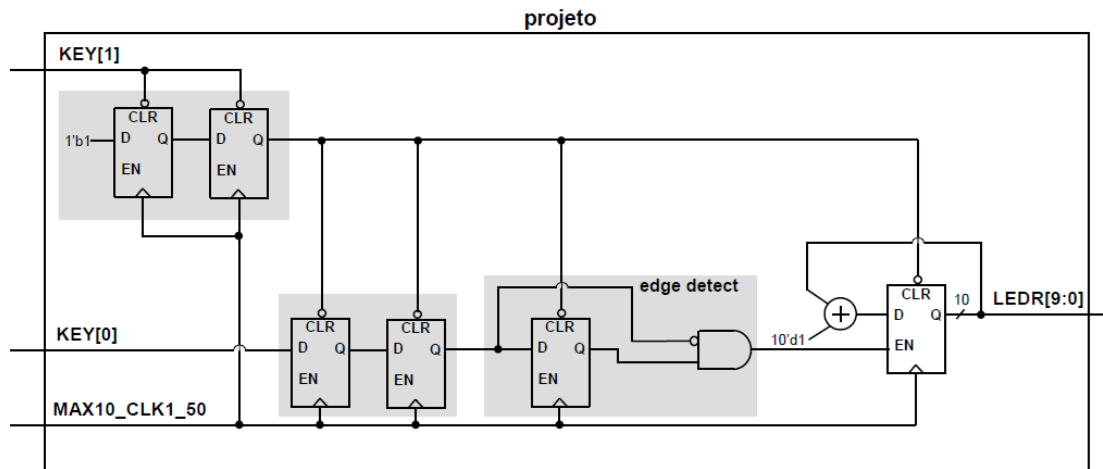
DIREITOS AUTORAIS

Este material didático é protegido por Direitos Autorais, sendo vedada a reprodução, distribuição, comercialização de qualquer material ou conteúdo dele obtido, sem a prévia e expressa autorização da Macnica DHW.

O material didático disponibilizado, se constitui em trabalho intelectual, cujos direitos autorais pertencem à Macnica DHW e a Intel, e está protegido por leis nacionais e tratados internacionais, qualquer reprodução ou distribuição, ou utilização comercial, de qualquer conteúdo deste material didático é expressamente proibida e o transgressor será responsabilizado civil e criminalmente na forma da lei.

PROJETO 6

- Abrir o projeto-06 e alterar o contador para 10 *bits*
- Conectar os 10 *bits* de saída diretamente nos 10 *LEDs*
- Adicionar um detector de borda para gerar um único pulso de *enable*
- Verificar o resultado no *SignalTap*



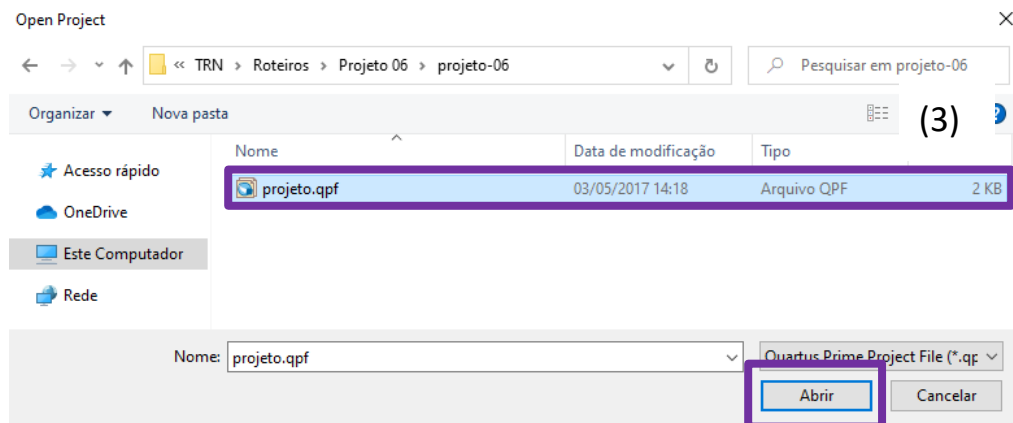
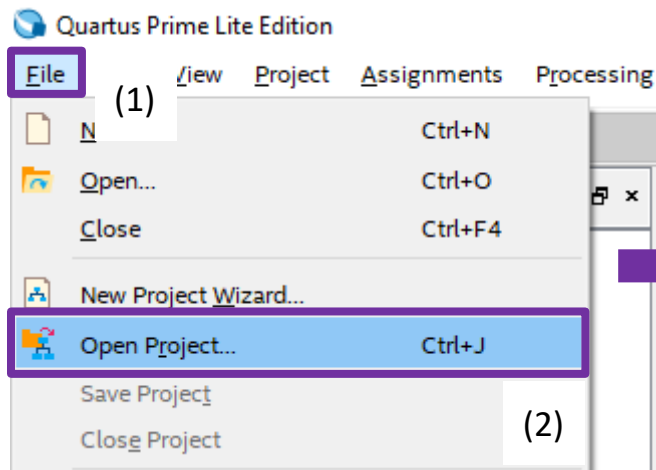
Para abrir o projeto 06

No Quartus:

Menu **File** -> **Open project** -> Vá a pasta Material do curso, caminho abaixo -> Abra o arquivo **projeto.qpf**

Caminho para o arquivo:

\TRN\Roteiros\Projeto 06\projeto-06



(4)

Para adicionar um detector de borda

Adicionando um detector de borda para gerar um único pulso de *enable*.

```
42 wire w_enablesinc_n = r_enablesinc2;
43
44 // contador
45
46 reg [31:0] r_contador;
47 always @ (posedge w_sysclk, negedge w_resetsinc_n) begin
48     if (~w_resetsinc_n) begin
49         r_contador <= 10'd0;
50     end else begin
51         if (w_enable_borda) begin
52             r_contador <= r_contador + 1;
53         end
54     end
55 end
56
57 // saída
58
59 assign LEDR = r_contador[31:22];
60
61 // detecta borda
62 reg r_enable_at;
63
64 always @(posedge w_sysclk, negedge w_resetsinc_n) begin
65     if (~w_resetsinc_n) begin
66         r_enable_at <= 1'b1;
67     end else begin
68         r_enable_at <= w_enablesinc_n;
69     end
70 end
71
72 wire w_enable_borda = r_enable_at & ~w_enablesinc_n;
73
74
75 endmodule
76
```

Para alterar o contador para 10 bits

```
42 wire w_enablesinc_n = r_enablesinc2;
43
44 // contador
45
46 reg [9:0] r_contador;
47 always @(posedge w_sysclk, negedge w_resetsinc_n) begin
48     if (~w_resetsinc_n) begin
49         r_contador <= 10'd0;
50     end else begin
51         if (w_enable_borda) begin
52             r_contador <= r_contador + 1;
53         end
54     end
55 end
56
57 // saida
58
59 assign LEDR = r_contador;
60
61 // detecta borda
62
63 reg r_enable_at;
64
65 always @(posedge w_sysclk, negedge w_resetsinc_n) begin
66     if (~w_resetsinc_n) begin
67         r_enable_at <= 1'b1;
68     end else begin
69         r_enable_at <= w_enablesinc_n;
70     end
71 end
72
73 wire w_enable_borda = r_enable_at & ~w_enablesinc_n;
74
75 endmodule
76
```

Alterando apenas em *reg* e *assign*.

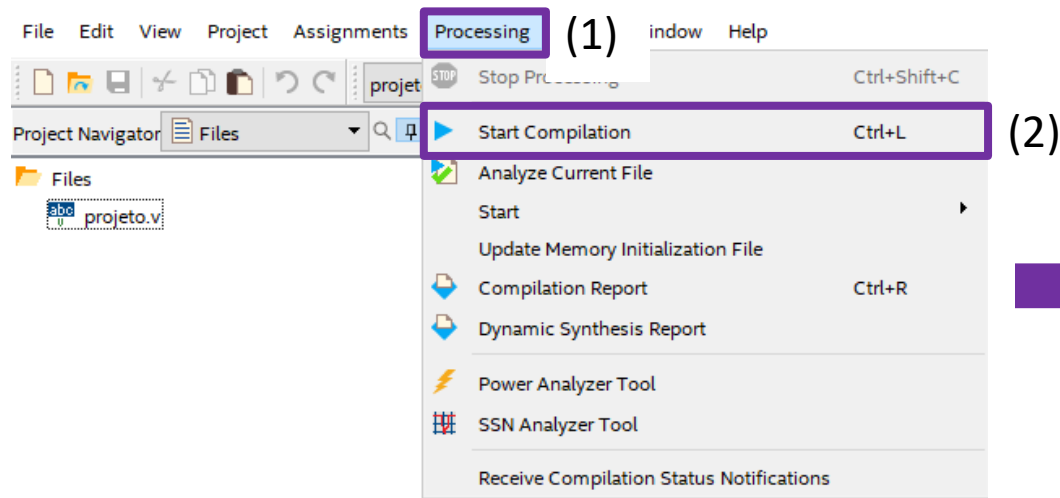
Verilog completo

```
20 r_resetsinc2 <= 1'b0;
21 end else begin
22   r_resetsinc1 <= 1'b1;
23   r_resetsinc2 <= r_resetsinc1;
24 end
25
26 wire w_resetsinc_n = r_resetsinc2;
27
28 // Sincroniza enable
29
30 reg r_enablesinc1, r_enablesinc2;
31 always @(posedge w_sysclk, negedge w_resetsinc_n) begin
32   if (~w_resetsinc_n) begin
33     r_enablesinc1 <= 1'b1;
34     r_enablesinc2 <= 1'b1;
35   end else begin
36     r_enablesinc1 <= w_enable_n;
37     r_enablesinc2 <= r_enablesinc1;
38   end
39 end
40
41 wire w_enablesinc_n = r_enablesinc2;
42
43 // Contador
44
45 reg [9:0] r_contador;
46 always @(posedge w_sysclk, negedge w_resetsinc_n) begin
47   if (~w_resetsinc_n) begin
48     r_contador <= 10'd0;
49   end else begin
50     if (w_enable_borda) begin // Altera o sinal enable
51       r_contador <= r_contador + 1;
52     end
53   end
54 end
55
56 // Saída
57
58 assign LEDR = r_contador;
59
60 //Detecta borda
61 reg r_enable_at;
62 always @(posedge w_sysclk, negedge w_resetsinc_n) begin
63   if (~w_resetsinc_n) begin
64     r_enable_at <= 1'b1;
65   end else begin
66     r_enable_at <= w_enablesinc_n;
67   end
68 end
69
70 wire w_enable_borda = r_enable_at & ~w_enablesinc_n;
71
72 endmodule
```

Para compilar o projeto

No Quartus:

Menu **Processing** -> **Start Compilation**



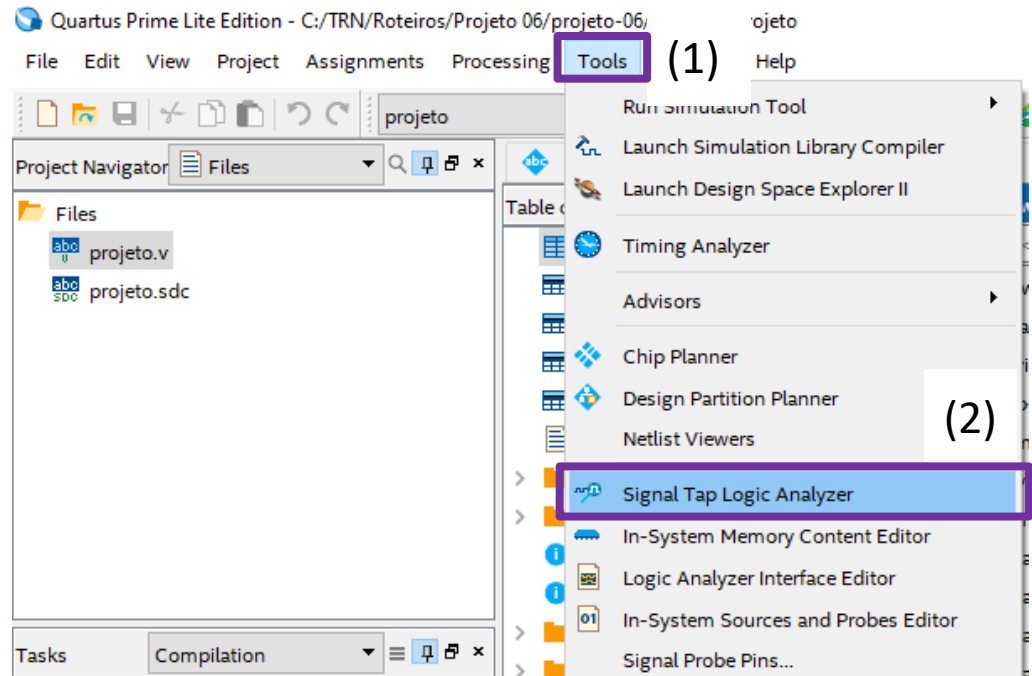
Tasks			Compilation	≡	🔍	×
	Task		Time			
✓	▼	▶ Compile Design	00:00:34			
✓	>	▶ Analysis & Synthesis	00:00:16			
✓	>	▶ Fitter (Place & Route)	00:00:10			
✓	>	▶ Assembler (Generate programming files)	00:00:04			
✓	>	▶ Timing Analysis	00:00:04			
	>	▶ EDA Netlist Writer				
		■ Edit Settings				
		▶ Program Device (Open Programmer)				

(3)

Para criar o *SignalTap*

No Quartus:

Menu Tools -> Signal Tap Logic Analyzer



Para configurar o *SignalTap*

Signal Tap Logic Analyzer - C:/TRN/Roteiros/Projeto 06/projeto-06/projeto - [stp1.stp]*

File Edit View Project Processing Tools Window Help

Instance Manager: Invalid JTAG configuration

Instance	Status	Enabled	LEs: 264	Memory: 0	Small: NA	Medium: NA	Large: NA
auto_signaltap_0	Not running	☒	264 cells	0 bits	NA	NA	NA

JTAG Chain Configuration: No device is selected

Hardware: Disabled Setup...

Device: None Detected Scan Chain

>> SOF Manager: [icon] [icon] [icon]

Node Finder

Named: *

Options

Filter: Signal Tap: pre-synthesis Customize...

Look in: [projeto] Include sub-hierarchy

Matching Nodes:

Name	Assignments
MAX10_CLK1_50	PIN_P11
r_enable_at	Unassigned
r_enableinc1	Unassigned
r_enableinc2	Unassigned
r_resetsinc1	Unassigned
r_resetsinc2	Unassigned
w_enable_borda	Unassigned
KEY	Unassigned
LEDR	Unassigned
SW	Unassigned
r_contador	Unassigned

Nodes Found:

Name	Assignments
MAX10_CLK1_50	PIN_P11

Signal Configuration:

Clock: MAX10_CLK1_50

Data

Sample depth: 1 K

☒ Segmented: 4 256 sample segments

Trigger Conditions

Seg Basic AND

Storage qualifier:

Type: Continuous

OK Cancel

0% 00:00:00

Intel

Para configurar o *SignalTap* (cont)

Signal Tap Logic Analyzer - C:/TRN/Roteiros/Projeto 06/projeto-06/projeto - projeto - [stp1.stp]*

File Edit View Project Processing Tools Window Help



Instance Manager: Invalid JTAG configuration

Instance	Status	Enabled	LEs: 264	Memory: 0	Small: NA	Medium: NA	Large: NA
auto_signaltap_0	Not running	☒	264 cells	0 bits	NA	NA	NA

Search altera.com

Node Finder

Named: *

Options

Filter: Signal Tap: pre-synthesis

Look in: [projeto]

Include sub-hierarchy view

Matching Nodes:	Nodes Found:
Name	Assignments
projeto	
MAX10_CLK1_50	PIN_P11
r_enable_at	Unassigned
r_enablesinc1	Unassigned
r_enablesinc2	Unassigned
r_resetsinc1	Unassigned
r_resetsinc2	Unassigned
w_enable_borda	Unassigned
KEY	Unassigned
LEDR	Unassigned
SW	Unassigned
r_contador	Unassigned

auto_signaltap_0 Lock mode: Allow all changes

Type	Alias	Name	Data Enable	Trigger Enable	Trigger Conditions
			0	0	Seg: Basic AND

Double-click to add nodes

Duplo clique na área branca da tela

Signal Configuration:

Clock: MAX10_CLK1_50

Data

Segmented: 4 256 samples

Nodes Allocated: Auto

Pipeline Factor: 0

Storage qualifier:

Type: Continuous

Hierarchy Display:

Data Log:

auto_signaltap_0

(4) List (2)

(3)

(6)

(5)

(7)

Insert Close



Para configurar o *SignalTap* (cont)

Clique com o botão direito do mouse em cima de **KEY[1..0]** e clique em **Ungroup** para desagrupar as **KEYS** e não serem consideradas como vetores.

auto_singaltap_0			Lock mode: Allow all changes		
Type	Alias	Name	Data Enable	Trigger Enable	Trigger Conditions
R		r_enable_at	15	15	Seg Basic AND
R		r_enablesinc2	15	15	Seg Basic AND
C		w_enable_borda	15	15	Seg Basic AND
L		KEY[1..0]	15	15	Xh
R		r_contador[9..0]	15	15	XXXh

×	Delete	Del
	Select All	Ctrl+A
	Find...	Ctrl+F
	Find Next	F3
	Find Bus Value...	Ctrl+Shift+F
	Find Next Bus Value	Ctrl+F3
	Find Previous Bus Value	Ctrl+Shift+F3
	Add Nodes with Plug-In	
	Add Nodes...	
	Plug-In Options...	
	Add State Machine Nodes...	
	Recreate State Machine Mnemonics...	
	Locate Node	
	Group	
	Ungroup	
	Ungroup Nested Groups	
	Rename	F2
	Mnemonic Table Setup...	
	Create Signal Tap List File	
	Invert Signal	
	Align Left	
	Align Right	
	Pre-Synthesis Node	
	Post-Fit Node	
	MSB on Top, LSB on Bottom	
	LSB on Top, MSB on Bottom	
	Bus Display Format	



Para configurar o *SignalTap* (cont)

Clique com o botão direito do mouse em cima do campo **Trigger Conditions** da chave **KEY[0]** e configure **Trigger** como borda de descida (**Falling Edge**).

The screenshot shows the 'auto_signaltap_0' configuration window. The 'Lock mode' is set to 'Allow all changes'. The table below lists the nodes and their trigger configurations. The 'KEY[0]' row is highlighted, and a context menu is open over its 'Trigger Conditions' column, showing the 'Falling Edge' option selected.

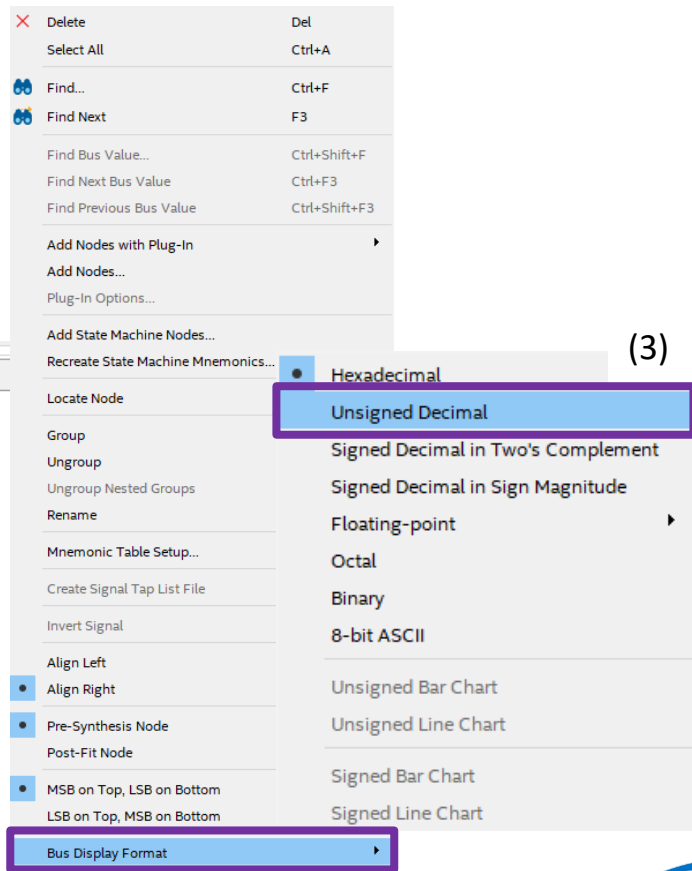
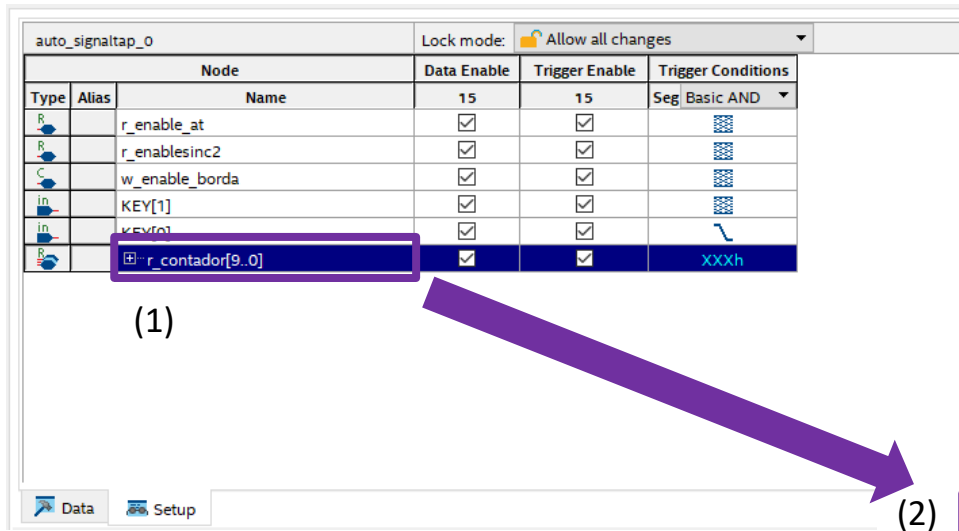
Type	Alias	Name	Data Enable	Trigger Enable	Trigger Conditions
R		r_enable_at	☒	☒	
R		r_enablesinc2	☒	☒	
C		w_enable_borda	☒	☒	
In		KEY[1]	☒	☒	
In		KEY[0]	☒	☒	
R		r_contador[9..0]	☒	☒	XXXh

(1)

(2)

Para configurar o *SignalTap* (cont)

Clique com o botão direito do mouse em cima do ***r_contador[9..0]*** para alterar o formato de amostragem do contador para decimal (*Unsigned Decimal*).



Para salvar as configurações feitas

No Signal Tap Logic Analyzer:

Menu **File** -> **Save** -> salve o arquivo .stp, caminho abaixo.

Caminho para o arquivo:

\TRN\Roteiros\Projeto 06\projeto-06\output_files

Observação:

O nome do arquivo pode ser de sua escolha. Clique em salvar, confirme novamente na tela seguinte para adicionar ao projeto.

(1) File menu open. (2) Save option highlighted. (3) Save dialog box showing the path. (4) File name 'depura_contador.stp' entered. (5) Save button highlighted. (6) Confirmation dialog box with Yes button highlighted.

Signal Tap Logic Analyzer - C:/TRN/Roteiros/Projeto 06/projeto-06/projeto - [stp1.stp]*

File Edit View Project Processing Tools Window Help

New File Ctrl+N

Close

Save Ctrl+S

Save As...

Create Signal Tap File

Export...

Page Setup...

Print Preview

Print... Ctrl+P

valid JTAG configuration

Enabled LEs: 780 Memory: 15360 Small: 0/0 Medium: 2/182 Large: 0/0

JTAG Chain Configuration: No device is selected

Hardware: Disabled Setup...

Device: None Detected Scan Chain

SOF Manager: [Icon]

Signal Configuration:

Clock: MAX10_CLK1_50

Data

Sample depth: 1 K RAM type: Auto

Segmented: 4 256 sample segments

Nodes Allocated: Auto Manual: 15

Pipeline Factor: 0

Storage qualifier:

Type: Continuous

Input port:

Salvar como

« TRN » Roteiros » Projeto 06 » projeto-06 » output_files

Pesquisar em output_files

Organizar Nova pasta

Quartus Prime

Do you want to enable Signal Tap File "output_files/depura_contador.stp" for the current project?

Yes No Cancel

Name: depura_contador.stp

Tip: Signal Tap Logic Analyzer Files (*.stp)

Add file to current project

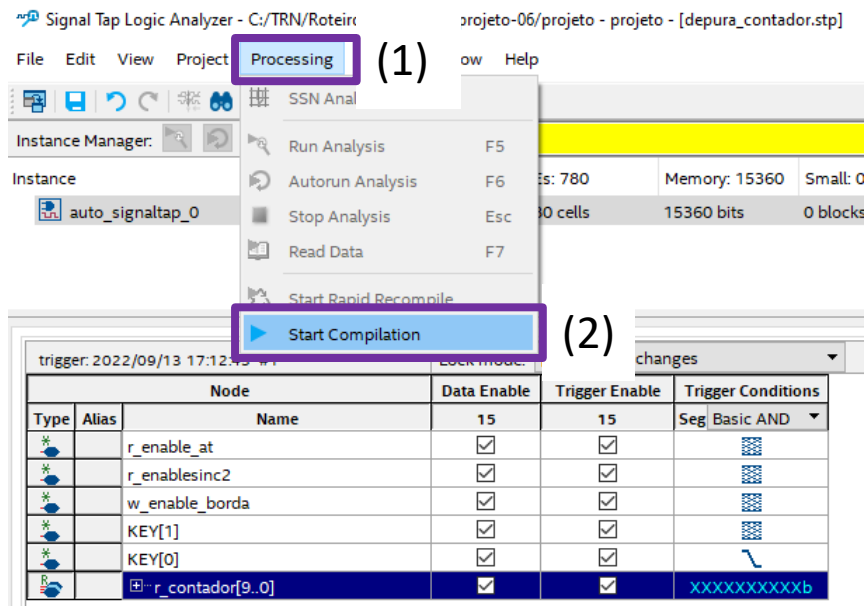
Salvar Cancelar



Para compilar o projeto novamente pelo *SinalTap*

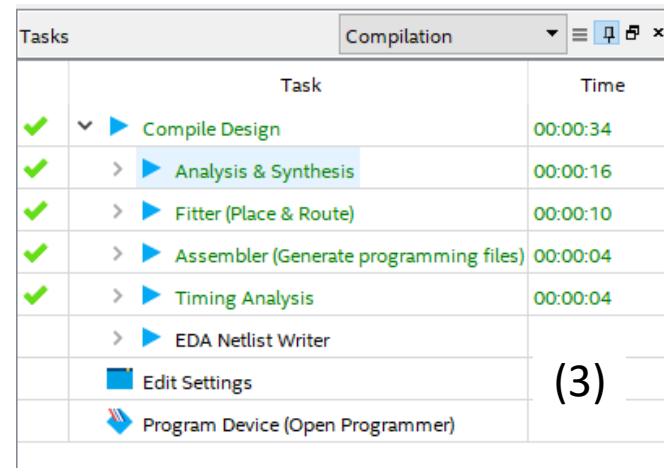
No Signal Tap Logic Analyzer:

Menu **Processing** -> **Start Compilation**



The screenshot shows the Signal Tap Logic Analyzer window. The 'Processing' menu is open, and the 'Start Compilation' option is highlighted. The window title is 'Signal Tap Logic Analyzer - C:/TRN/Roteir...'. The menu items are: Run Analysis (F5), Autorun Analysis (F6), Stop Analysis (Esc), Read Data (F7), and Start Rapid Recompile. The 'Start Compilation' option is highlighted with a blue box and labeled (2). The background shows a table with columns: Type, Alias, Name, Data Enable, Trigger Enable, and Trigger Conditions. The table contains several rows, including 'r_enable_at', 'r_enablesinc2', 'w_enable_borda', 'KEY[1]', 'KEY[0]', and 'r_contador[9..0]'.

Type	Alias	Name	Data Enable	Trigger Enable	Trigger Conditions
*		r_enable_at	✓	✓	
*		r_enablesinc2	✓	✓	
*		w_enable_borda	✓	✓	
*		KEY[1]	✓	✓	
*		KEY[0]	✓	✓	
*		r_contador[9..0]	✓	✓	XXXXXXXXXXb



The screenshot shows the 'Tasks' window with the 'Compilation' tab selected. The table lists the tasks and their completion status. The 'Compile Design' task is highlighted with a blue box and labeled (3). The tasks are: Compile Design (00:00:34), Analysis & Synthesis (00:00:16), Fitter (Place & Route) (00:00:10), Assembler (Generate programming files) (00:00:04), Timing Analysis (00:00:04), EDA Netlist Writer, Edit Settings, and Program Device (Open Programmer).

Task	Time
✓ Compile Design	00:00:34
✓ > Analysis & Synthesis	00:00:16
✓ > Fitter (Place & Route)	00:00:10
✓ > Assembler (Generate programming files)	00:00:04
✓ > Timing Analysis	00:00:04
> EDA Netlist Writer	
Edit Settings	
Program Device (Open Programmer)	

Para enviar ao Kit

Signal Tap Logic Analyzer - C:/TRN/Roteiros/Projeto 06/projeto-06/projeto - projeto - [depura_contador.stp]

File Edit View Project Processing Tools Window Help

The screenshot shows the Signal Tap Logic Analyzer interface. The 'Instance Manager' pane on the left lists an instance named 'auto_signaltap_0' with a status of 'Not running'. The main area displays an 'Invalid JTAG configuration' error. The 'JTAG Chain Configuration' dialog is open, showing 'No device is selected' in a red box. The 'Hardware' dropdown is set to 'Disabled' and the 'Device' dropdown is set to 'None Detected'. The 'SOF Manager' pane shows a file named 'output_files/projeto.sof'.

Instance	Status	Enabled	LEs: 780	Memory: 15360	Small: 0/0	Medium: 2/182	Large: 0/0
auto_signaltap_0	Not running	☒	780 cells	15360 bits	0 blocks	2 blocks	0 blocks

No Signal Tap Logic Analyzer:

Selecione o arquivo **projeto.sof** no campo **JTAG Chain Configuration** -> ... e utilize o ícone **Program Device** para gravar no Kit usando a **USB-Blaster**

Caminho para o arquivo:

\\TRN\\Roteiros\\Projeto 06\\projeto-06\\output_files

The 'Select Programming File' dialog box is shown. The 'Look in' field displays the path 'C:/TRN/Roteiros/Projeto 06/projeto-06/output_files'. The file 'projeto.sof' is selected in the file list. The 'File name' field contains 'projeto.sof'. The 'Files of type' dropdown is set to 'SRAM Object Files (*.sof)'. The 'Open' button is highlighted.

Para verificar o resultado no *SignalTap*

Signal Tap Logic Analyzer - C:/TRN/Roteiros/Projeto 06/projeto-06/projeto - projeto - [depura_contador.stp]

File Edit View Project Processing Tools Window Help

Instance Manager: Invalid JTAG configuration

Instance	Status	Enabled	LEs: 780	Memory: 15360	Small: 0/0	Medium: 2/182	Large: 0/0
auto_signaltap_0	Not running	☒	780 cells	15360 bits	0 blocks	2 blocks	0 blocks

JTAG Chain Configuration: No device is selected

Hardware: Disabled Setup...

Device: None Detected Scan Chain

>> SOF Manager: output_files/projeto.sof ...

trigger: 2022/09/13 17:12:43 #1 Lock mode: Allow all changes

Type	Alias	Name	Data Enable	Trigger Enable	Trigger Conditions
*		r_enable_at	☒	☒	Seg: Basic AND
*		r_enablesinc2	☒	☒	
*		w_enable_borda	☒	☒	
*		KEY[1]	☒	☒	
*		KEY[0]	☒	☒	
*		r_contador[9..0]	☒	☒	XXXXXXXXXXb

Signal Configuration:

Clock: MAX10_CLK1_50

Data

Sample depth: 1 K RAM type: Auto

☒ Segmented: 4 256 sample segments

Nodes Allocated: ☒ Auto ☐ Manual: 15

Hierarchy Display: projeto

Data Log: auto_signaltap_0

auto_signaltap_0

100% 00:00:43

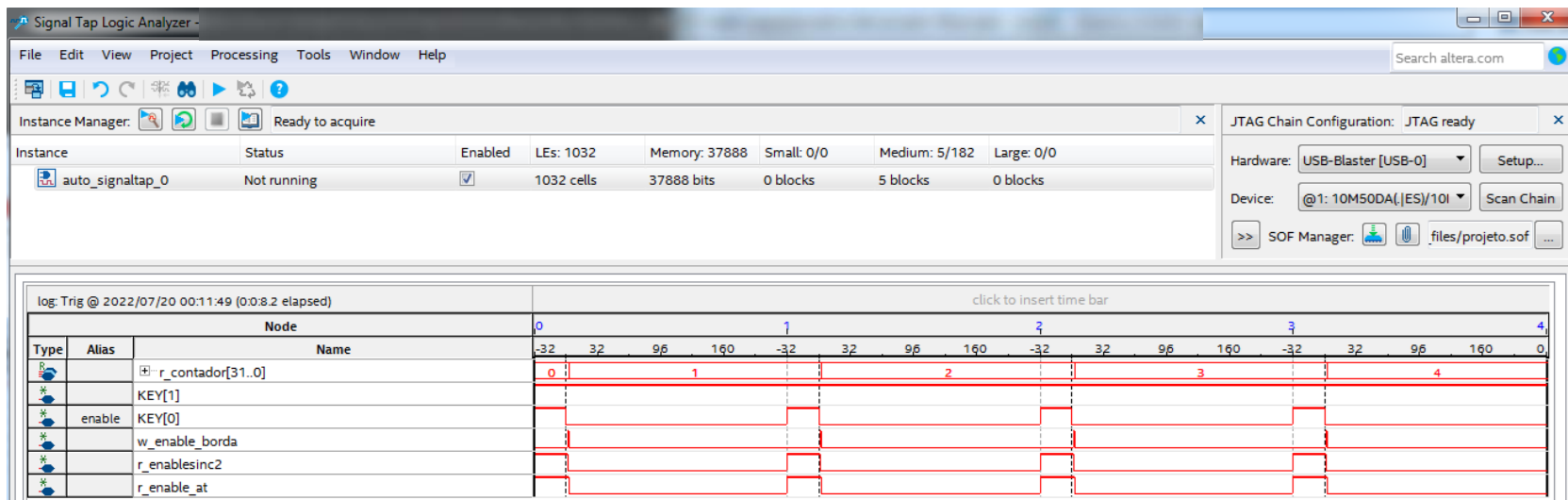
No Signal Tap Logic Analyzer:

Clique no ícone **Run Analysis** no canto superior esquerdo

Observação:

O Signal Tap aguarda por 4 segmentos de trigger do botão **KEY[0]**, sendo necessário acionar o botão no Kit.

Para verificar o resultado no *SignalTap* (cont)



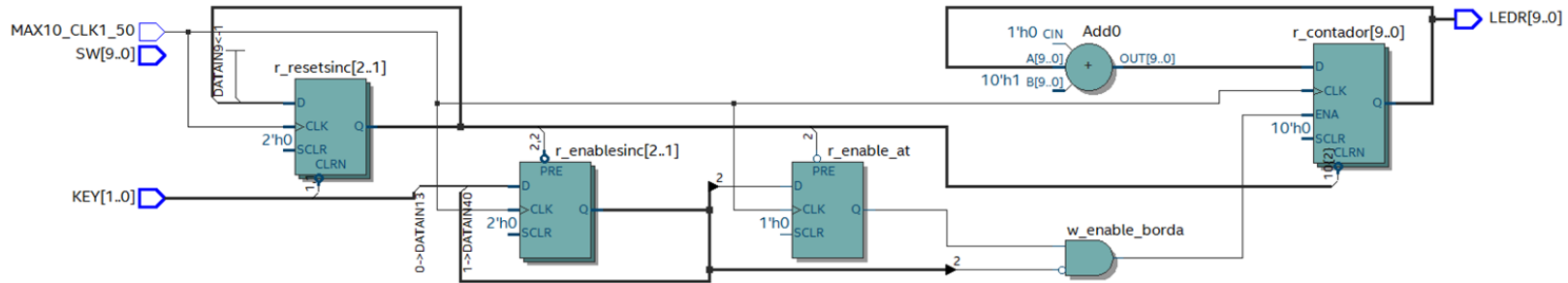
Observação:

Podemos visualizar todo *Buffer* de execução

Para visualizar o esquemático no RTL Viewer

No Quartus:

Menu **Tools** -> **Netlist Viewers** -> **RTL Viewer**



Visualização mais próxima
ao pedido do laboratório.

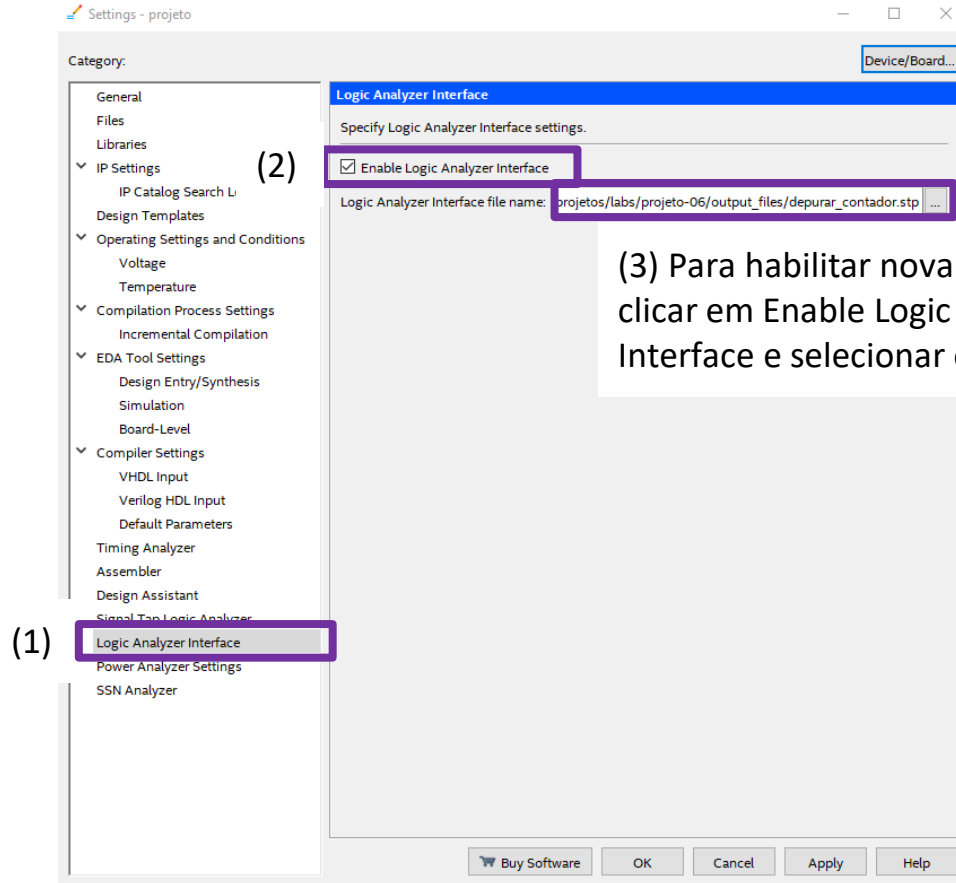
Para desabilitar/habilitar Signal Tap no projeto

No Quartus:

Menu **Assignments** ->

Settings ->

Signal Tap Logic Analyzer



(3) Para habilitar novamente basta clicar em Enable Logic Analyzer Interface e selecionar o arquivo .stp

Para testar no Kit DE10-Lite

A KEY[0] realiza a incrementação com detecção de borda dos 10 leds, enquanto a KEY[1] realiza o reset, voltando os leds ao estado padrão.

KEY[0]	KEY[1]	LEDR[9:0]
Pressionada	Não Pressionada	Incrementa
Indiferente	Pressionada	Reset



Obrigado

